MILITARY SPECIFICATION

ADAPTER, LAUNCHER, AIRCRAFT, ADU-299/E

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification establishes the requirements for the manufacture, inspection and testing of an aircraft missile launcher adapter that can be suspended from various bomb ejector racks, as noted hereinafter.

1.2 Classification - The equipment covered by this specification shall be of one type designated as ADU-299/E, Adapter, Launcher, Aircraft, and hereinafter referred to as Adapter.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-6021	Castings, classification and inspection of
MIL-T-7743	Testing, store suspension equipment
MIL-N-18307	Nomenclature and nameplates
MIL-A-21180	Aluminum-Alloy castings, high strength
MIL-L-22620	Inspection system requirements

STANDARDS

Military

MIL-STD-129	Marking for shipment and storage
MIL-STD-130	Identification marking of U.S. Military Property
MIL-STD-143	Specifications and standards, order of precedence for the selection of

THIS DOCUMENT CONTAINS 7 PAGES. FSC 1055

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Continued)

STANDARDS

Military

MIL-STD-794	Parts and equipment, procedures for packaging and packing of

Bureau of Naval Weapons

DL67A71A1	List of Drawings - Adapter, Launcher, Aircraft, ADU-299/E

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Parts and Material - Parts and materials used in the manufacture of the adapter shall conform to the applicable specifications as specified herein and on drawing DL67A71A1. Substitution of parts and materials may be permitted provided they are equivalent to the original Government specifications, are approved by the procuring activity in writing, and contain requirements for adequate material tests. The use of any substitute part or material shall not constitute a waiver of Government inspection.

3.1.1 Selection of Materials - Specifications and Standards of all materials and parts not specifically designated herein, shall be selected in accordance with MIL-STD-143.

3.2 Design and Contruction -

3.2.1 General - Construction, dimensions and finishes for the adapter shall be in accordance with the applicable drawings listed under DL 67A71A1.

3.3 Reliability - The adapter shall pass the inspection requirements of Section 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

3.4 Performance Characteristics - The adapter shall be capable of being suspended from each of the Aero 7A, Aero 3A, MAU9/A, and MAU9A/A Ejector Bomb Racks using either the 14 inch or 30 inch suspension lugs. With any rack suspension used, the adapter shall permit either an LAU-7A or Aero 3A guided missile launcher to be mounted to its bottom surface. Interconnection of adapter cables and launcher cables shall be made manually.

3.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.6 Workmanship - All details of workmanship on the adapter shall be uniform and of a high level of quality. All parts shall be free from excessive marks, irregularities, or defects which could adversely affect performance, reliability, maintainability or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract, the contractor will be responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections and tests required by this Specification when deemed necessary to assure that the adapters conform to prescribed requirements.

4.2 Classification of Tests - The tests on the adapters shall be classified as follows:

a. Preproduction Tests

b. Quality Conformance Inspection

4.3 Preproduction Tests - Preproduction tests shall consist of all the tests of this specification.

4.3.1 The contractor shall furnish two preproduction adapters fabricated by quantity production methods and tools as soon as practicable after award of contract or order. Both adapters shall be delivered for preproduction tests to a Government specified testing laboratory. The order of conducting the tests shall be at the discretion of the testing laboratory unless otherwise specified by the procuring activity. Quantity production shall not start until the preproduction samples have been found satisfactory by the Government procuring agency. When a contractor is in continuous production of adapters from contract to contract, submission of additional preproduction adapters may be waived at the discretion of the Government procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection - This inspection shall consist of the individual and the sampling tests. All sampling tests will be performed by a government specified laboratory. (See 6.2)

4.4.1 Individual Tests - Each adapter on contract or order shall be subjected to the following tests:

	Paragraph
a. Examination of Adapter	4.5.2
b. Operational Tests	4.5.3
c. Radiographic Inspection (Adapter Casting)	4.5.1

4.4.2 Sampling Tests - A number of adapters shall be selected from each lot in accordance with the following table:

TABLE I

SAMPLE SCHEDULE

LOT SIZE	SAMPLE ADAPTERS REQUIRED
99 or less	2
100 to 300	3

The samples shall have passed the individual tests and shall be subjected to the following inspections and tests. If a sample fails to pass any of these tests, the lot represented by that sample shall be rejected.

TABLE II

SAMPLING TESTS

ADAPTER TESTS	PARAGRAPH	SAMPLE NO.
Physical Characteristics	4.5.4	1 & 2
Electrical Characteristics	4.5.4	1
Vibration - Resonance Only	4.5.4	2
Shock	4.5.4	1
Static	4.5.5	2 (or 3) *
Insulation	4.5.4	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

*NOTE: When only two samples are available, samples for static test will be same as for vibration test, repair as required. If an additional sample is available, it will be used for the static test.

4.5 Test Methods -

4.5.1 Radiographic Inspection - The adapter castings shall be subjected to Radiographic inspection in accordance with Specification MIL-A-21180, Class 1, and MIL-C-6021, Class 1B.

4.5.2 Examination - The adapter shall be thoroughly examined to insure it has been properly assembled and that it conforms with this specification and applicable drawings listed in 2.1 with respect to all the requirements of material, workmanship, and operational adequacy.

4.5.3 Operational Test - With either the 14 inch or 30 inch bomb lugs inserted in the adapter, the adapter shall latch securely to any 14 inch or 30 inch suspension bomb rack listed in 3.4. Both 3/4-16 adapter nuts shall permit the 30 inch spaced mounting bolts on the LAU-7/A or the Aero 3A launchers to be tightened against the bottom of the adapter. During attachment of launcher to adapter, cables between the adapter and launcher shall be manually interconnected with their corresponding mating connectos. All adapter cable wires shall be tested for continuity between connector pins as shown on the applicable drawings listed in 2.1.

4.5.4 Adapter Tests - The adapter shall be subjected to and pass the following tests specified in Specification MIL-T-7743:

a. Physical Characteristics

b. Electrical Characteristics

c. Vibration - Resonance Only

d. Shock

e. Insulation

4.5.5 Static Test - Static load tests will be conducted with the adapter suspended from a MAU-9A/A rack or simulated fixture. An LAU-7A launcher/sidewinder 1C (AIM-9D) missile combination or equivalent simulated configuration having the same center of gravity location shall be mounted to the adapter. The static load tests of paragraph 4.5.9 of Specification MIL-L-22620 shall be applied in 10% increments at the center of gravity of the launcher/missile combination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

4.6 Rejections -

4.6.1 Individual Rejection - The failure of any adapter or part thereof to pass the individual inspections outlined in 4.4.1 shall cause its rejection.

4.6.2 Sampling Rejection - The failure of any lot sampling to pass the sampling tests in 4.4.2 shall cause rejection of all adapters in that lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - The adapter shall be preserved and packaged in accordance with Specification MIL-STD-794, Level B.

5.2 Packing - The adapter shall be packed in accordance with Specification MIL-STD-794, Level B.

5.3 Marking - Marking shall conform to Standard MIL-STD-129.

6. NOTES

6.1 Intended Use - Suspension of the adapter from an aircraft ejector bomb rack, permits either an LAU-7A or Aero 3A launcher to be mounted from that same bomb rack station.

6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification

b. If preproduction test is required (See 4.3.1)

c. Where the preproduction test is to be conducted

d. Any adapter samples sent for preproduction or sampling tests are not to be considered as part of the contract order.

6.3 Availability of Bomb Rack for Tests - The contractor may submit a request to the procuring agency for an ejector bomb rack to be used for the operational fit tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.

6.4 Precedence of Documents - In case of conflict between this specification, the contract, or other applicable documents, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification or drawing.

b. This Specification - This specification shall have precedence over all other applicable specifications and drawings. Any deviation from this specification or other specifications and drawings, shall be approved in writing by the procuring agency.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:00Z
Check the source to verify that this is the current version before use.